Sayan Biswas

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Research interests

Trustworthy Collaborative and Decentralized ML (focusing on Privacy, Adversarial Robustness, and Fairness)

Positions

2023 - present **Postdoctoral Researcher** at EPFL (Scalable Computing Systems Lab) – Lausanne, Switzerland.
Supervisor: Prof. Anne-Marie Kermarrec.

Education

2020 - 2023 **PhD in Computer Science** from INRIA, École Polytechnique, and Institut Polytechnique de Paris – Palaiseau, France.
Supervisor: Prof. Catuscia Palamidessi.

2016 - 2020 **Master of Mathematics (M.Math.)** with **First-Class Honours** from University of Bath – Bath, England.

Grants and Awards

2025 **Young Scholars' Development Program Fellowship** from ACM CCS'25.

2025 **Grant of CHF 65,452.00 from the Federal Office for Defence Procurement of Switzerland and *armasuisse*** for the project "Privacy-preserving and distributed processing of public data in hybrid trust networks".

2024 **Award for the Best PhD Thesis in Computer Science** from Institut Polytechnique de Paris, France.

Honors and Recognition

2026 **Invited participant** at **Dagstuhl Seminar 26401 on "Divided We Stand: Security and Privacy of Collaborative Learning"** held at Schloss Dagstuhl - Leibniz Center for Informatics in Wadern, Germany.

Publications

Peer-Reviewed Conferences

2026 ***Federated and differentially private estimation of KL divergence*** by *Sayan Biswas*, Graham Cormode, Carsten Maple, and Mary Scott, at *IEEE International Conference on Big Data (Big Data) 2026*.

2026 ***Your Neighbors Know: Leveraging Local Neighborhoods for Backdoor Detection in Decentralized Learning*** by *Sayan Biswas*, Antoine Boutet, Davide Frey, Romaric Gaudel, Rachid Guerraoui, Maxime Jacovella, Anne-Marie Kermarrec, Dimitri Lerévérend, Francois Taiani, and Martijn de Vos, at *Advances in Neural Information Processing Systems (NeurIPS) 2026*.

2026 ***Communication-Efficient Secure Aggregation in Decentralized Learning*** by *Sayan Biswas*, Anne-Marie Kermarrec, Rafael Pires, Rishi Sharma, and Milos Vujasinovic, at *International Symposium on Reliable Distributed Systems (SRDS) 2026*.

2025 ***Mitigating Membership Inference Vulnerability in Iterative Federated Clustering Algorithm*** by Kangsoo Jung, *Sayan Biswas*, and Catuscia Palamidessi at *Workshop on Recent Advances in Resilient and Trustworthy Machine Learning-Driven Systems (ARTMAN) co-located with ACM Conference on Computer and Communications Security (CCS) 2025*.

2025 ***Robust ML Auditing using Prior Knowledge*** by Jade Garcia Bourrée, Augustin Godinot, *Sayan Biswas*, Anne-Marie Kermarrec, Erwan Le Merrer, Gilles Tredan, Martijn de Vos, and Milos Vujasinovic at *International Conference on Machine Learning (ICML) 2025*. **Accepted as *spotlight* (top** 2.6% **of the papers)**.

2025 ***Low-Cost Privacy-Aware Decentralized Learning*** by *Sayan Biswas*, Davide Frey, Romaric Gaudel, Anne-Marie Kermarrec, Dimitri Lerévérend, Rafael Pires, Rishi Sharma, and François Taïani at *Privacy Enhancing Technologies Symposium (PoPETs) 2025*.

2025 ***Boosting Asynchronous Decentralized Learning with Model Fragmentation*** by *Sayan Biswas*, Anne-Marie Kermarrec, Alexis Marouani, Rafael Pires, Rishi Sharma, and Martijn de Vos at *The ACM Web Conference (WWW) 2025*. **Selected for *oral presentation* (top** 7% **of the papers)**.

2024 ***Fair Decentralized Learning*** by *Sayan Biswas*, Anne-Marie Kermarrec, Rishi Sharma, Thibaud Trinca, and Martijn de Vos at *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML) 2025*.

2024 ***Noiseless Privacy-Preserving Decentralized Learning*** by *Sayan Biswas*, Mathieu Even, Anne-Marie Kermarrec, Laurent Massoulie, Rafael Pires, Rishi Sharma, and Martijn de Vos at *Privacy Enhancing Technologies Symposium (PoPETs) 2025*.

2023 ***Tight Differential Privacy Guarantees for the Shuffle Model with*** k***-Randomized Response*** by *Sayan Biswas*, Kangsoo Jung, and Catuscia Palamidessi, at *International Symposium on Foundations and Practice of Security (FPS) 2023*.

2023 ***PRIVIC: A privacy-preserving method for incremental collection of location data*** by *Sayan Biswas* and Catuscia Palamidessi, at *Privacy Enhancing Technologies Symposium (PoPETs) 2024*.

2023 ***Group privacy for Personalized Federated Learning*** by Filippo Galli, *Sayan Biswas*, Kangsoo Jung, Tommaso Cucinotta, and Catuscia Palamidessi, at *International Conference on Information Systems Security and Privacy (ICISSP) 2023*.

2022 ***Impact of Sampling on Locally Differentially Private Data Collection*** by *Sayan Biswas*, Graham Cormode, and Carsten Maple, at *Competitive Advantage in the Digital Economy – Resilience, Sustainability, Responsibility, and Identity (CADE) 2022*. **Winner of the Best Paper Award**.

2022 ***Tight Differential Privacy Blanket for Shuffle Model*** by *Sayan Biswas*, Kangsoo Jung, and Catuscia Palamidessi, at *Competitive Advantage in the Digital Economy – Resilience, Sustainability, Responsibility, and Identity (CADE) 2022*.

2021 ***An Incentive Mechanism for Trading Personal Data in Data Markets*** by *Sayan Biswas*, Kangsoo Jung, and Catuscia Palamidessi, at *International Colloquium on Theoretical Aspects of Computing (ICTAC) 2021*.

Journals

2026 ***Practical and Private Hybrid ML Inference with Fully Homomorphic Encryption*** by *Sayan Biswas*, Philippe Chartier, Akash Dhasade, Tom Jurien de la Gravière, David Kerriou, Anne-Marie Kermarrec, Mohammed Lemou, Franklin Tranie, Martijn de Vos, and Milos Vujasinovic. *Transactions on Machine Learning Research (2026) (to appear)*.

2024 ***A Privacy-Preserving Querying Mechanism with High Utility for Electric Vehicles*** by Ugur Ilker Atmaca, *Sayan Biswas*, Carsten Maple, and Catuscia Palamidessi. *IEEE Open Journal of Vehicular Technology, Volume 5 (2024)*.

2023 ***Advancing Personalized Federated Learning: Group Privacy, Fairness, and Beyond*** by Filippo Galli, Kangsoo Jung, *Sayan Biswas*, Catuscia Palamidessi, and Tommaso Cucinotta. *Springer Nature Computer Science, Volume 4, Issue 6, Article 831 (2023)*.

Book Sections

2021 ***Establishing the Price of Privacy in Federated Data Trading*** by Kangsoo Jung, *Sayan Biswas*, and Catuscia Palamidessi. *Protocols, Strands, and Logic, pp 232-250, LNCS 13066, Springer*.

Non-Archival Workshops

2025 ***Robust ML Auditing using Prior Knowledge*** by Jade Garcia Bourrée, Augustin Godinot, *Sayan Biswas*, Anne-Marie Kermarrec, Erwan Le Merrer, Gilles Tredan, Martijn de Vos, and Milos Vujasinovic, at *Technical AI Governance Workshop in conjunction with ICML (TAIG ICML) 2025*.

2024 ***Bayes' capacity as a measure for reconstruction attacks in federated learning*** by *Sayan Biswas*, Mark Dras, Pedro Faustini, Natasha Fernandes, Annabelle McIver, Catuscia Palamidessi, and Parastoo Sadeghi, at *Workshop on Security, Privacy, and Information Theory (Protect-IT) in conjunction with IEEE Computer Security Foundations Symposium (CSF) 2024*.

2023 ***PRIVIC: A privacy-preserving method for incremental collection of location data*** by *Sayan Biswas* and Catuscia Palamidessi at *Theory and Practice of Differential Privacy Workshop (TPDP) 2023*.

2023 ***On the adaptive sensitivity of differentially private machine learning*** by Filippo Galli, *Sayan Biswas*, Kangsoo Jung, Tommaso Cucinotta, and Catuscia Palamidessi, at *Privacy-Preserving Artificial Intelligence (PPAI) in conjunction with AAAI 2023*.

2022 ***Group privacy for personalized federated learning*** by Filippo Galli, *Sayan Biswas*, Kangsoo Jung, Tommaso Cucinotta, and Catuscia Palamidessi, at *International Workshop on Federated Learning: Recent Advances and New Challenges (FL-NeurIPS) in conjunction with NeurIPS 2022*. **Accepted for *oral presentation* (top 10% of the papers)**.

Teaching

Aug'22 - Sep'22 **INRIA-DFKI European Summer School on AI (IDESSAI 2022)**, *Saarbrücken, Germany*,
Differential Privacy & Federated Learning: theory and implementation (voted as the *most liked course* of IDESSAI 2022).

Feb'21 - Jun'23 **École Polytechnique**, *Palaiseau, France*,
CSE 102: Advanced Programming with Python.

Oct'18 - May'20 **University of Bath**, *Bath, England*,
XX10190: Programming & Discrete Mathematics with MATLAB (2018-19), MA10209: Algebra 1A (2019-20), and MA10212: Prob. & Stat. 1B (2019-20) .

Jul'17 - Aug'17 **Humen Foreign Language School (HFLS)**, *Humen, China*,
Taught Mathematics and English at the HFLS International Summer Camp.

Academic Services

PC member IEEE ICDCS 2026, AISTATS 2026, ACM CCS 2026, IEEE SaTML 2026, ESORICS Workshop on Secure and Trustworthy Machine Unlearning Systems 2025, IEEE SaTML 2025, AAAI Workshop Privacy Preserving Artificial Intelligence 2023

Ad hoc peer reviewer ACM Transactions on Privacy and Security, IEEE Journal on Selected Areas in Information Theory, IEEE Transactions on Dependable and Secure Computing

Select Talks

2026 ***Decentralised ML: Trust No One, Train Together***. *Foundations of AI Security and Privacy Workshop at Swiss AI Days 2026*. March 25, 2026. Fribourg, Switzerland.

2022 ***Incremental collection of location data: Differential Privacy and beyond...***. *7th Franco-Japanese Cybersecurity Workshop 2022*. October 24, 2022. Tokyo, Japan.

Miscellaneous Research Appointments

Sep'23 - Oct'23 **Visiting Scholar**, *Macquarie University*, Sydney, Australia. Hosted by Prof. Annabelle McIver and Dr. Natasha Fernandes.

Jan'22 - Mar'22 **Visiting Scholar**, *The University of Warwick*, Coventry, UK. Hosted by Prof. Carsten Maple and Prof. Graham Cormode.

Jun'20 - Aug'20 **Research Intern**, *The University of Warwick*, Coventry, UK. Supervised by Prof. Graham Cormode and Prof. Carsten Maple.

Jun'19 - Sep'19 **Research Intern**, *INRIA*, Palaiseau, France. Supervised by Prof. Catuscia Palamidessi.

Jun'18 - Aug'18 **Research Intern**, *Institute for Mathematical Innovation & University of Bath*, Bath, UK. Supervisor: Prof. Christopher Jennison.

Other Achievements

2022 Winner of the **Best Paper Award** at the conference on Competitive Advantage in the Digital Economy (CADE) 2022.

2018 **Qualified for the International Collegiate Programming Contest (ICPC) European Finals 2018** *(first solver of Problem 5 in the UK and Ireland qualification round)* representing University of Bath.

2016 **Honourable Mention in the final round of Indian National Philosophy Olympiad 2016** *(top 10 from India)*.

2016 **Qualified the Zonal Informatics Olympiad 2016** *(top 6 in West Bengal, India)*.

2015 **Honourable Mention at the International Linguistics Olympiad Training Camp 2015** *(the invitational round for Indian team selection for IOL 2015) (top 15 in India)*.

2013 & 2014 **Qualified for the Regional Mathematical Olympiad 2013 & 2014** *(top 28 in West Bengal, India)*.

Other Work Experience

Nov'18 - May'20 Lead student-editor of Dept. of Mathematics newsletter of University of Bath

Jul'19 Invigilator and examiner for the International Mathematical Olympiad (IMO) 2019 at Bath, UK

Feb'17 - Nov'19 Project coordinator and head for Mathscon (UK's largest student-led maths conference)

Jan'17 - Mar'20 Volunteer for United Kingdom Mathematics Trust and Mentor for British Mathematical Olympiad